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17.

$C + O_2 \rightarrow CO_2$ $\quad \Delta H^0 = -$ a KJ ..... (i)

$2CO + O_2\ 2CO_2$ $\quad \Delta H^0 = -$b kJ ..... (ii)

$$C + \frac{1}{2}O_2 \rightarrow CO\ \Delta H^0 = ?$$

(i) × 2

$2C + 2O_2 \rightarrow 2CO_2$ $\quad \Delta H^0 = -$ 2a kJ ..... (iii)

Reverse of equation (ii) will be

$2CO_2 \rightarrow 2CO + O_2$ $\quad \Delta H^0 = +$ b kJ ..... (iv)

(iii) + (iv)

$2C + O_2 \rightarrow 2CO\ \Delta H^0 = b - 2a$ kJ ..... (v)

(v) ÷ 2

$$C + O_2 \rightarrow CO\ \Delta H^0 = \frac{(b-2a)}{2}\ kJ$$

18.

Given :

$\Delta H_C\ (CH_4)$ $\quad = -$ 890 kJ mol^{-1}

$\Delta H_C\ (C_3H_8)$ $\quad = -$ 2220 kJ mol^{-1}

Let the mixture contain lit of and lit of propane.

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

x $\qquad$ 2x

$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$

(15.68 – x) $\qquad$ 5 (15.68 – x)

Volume of oxygen consumed = 2x + 5 (15.68 – x) = 32 lit

2x + 78.4 – 5x = 32 l

78.4 – 3x = 32

3x = 46.4 l

x = 15.47 l